

Sujay Seeram

Electronics and Communication Engineer | Cyber-Physical Systems & Embedded Hardware

Email: sujayat2007@gmail.com | GitHub: github.com/seeramsujay | Portfolio: seeramsujay.github.io

Location: Coimbatore, Tamil Nadu / Andhra Pradesh, India

ENGINEERING PROFILE & CORE PHILOSOPHY

Electronics and Communication Engineering undergraduate at Amrita Vishwa Vidyapeetham specializing in **resource-constrained cyber-physical systems**, **embedded hardware architecture**, and **physics-informed computing**. Pioneer of the “**Extreme Economic Engineering**” paradigm: deriving hardware configurations from physical first principles to drastically slash physical component counts, quiescent power consumption, and bill-of-materials (BOM) while maximizing system capability and sensor fidelity. Disciplined NCC cadet, 3-time Services Selection Board (SSB) attendee, and certified in Indian Sign Language (ISL).

EDUCATION

- **Amrita Vishwa Vidyapeetham** Coimbatore, India
Bachelor of Technology in Electronics and Communication Engineering (ECE) *Expected 2027*
 - **Core Coursework:** Microprocessors & Microcontrollers (ARM Cortex/8051), Signals & Systems, Control Systems, Digital System Design, Analog Electronic Circuits, VLSI Design, Electromagnetic Theory, Communication Networks, Power Electronics.
 - **Academic Focus:** Cyber-Physical Architectures, Acoustic/Vibration Signal Processing, Embedded Edge ML, Sensor Conditioning.

TECHNICAL SKILLS

- **Silicon, SoCs & MCUs:** STM32 (ARM Cortex-M4), ESP32 (Dual-Core Xtensa, RISC-V ULP), Raspberry Pi, RP2040, ATmega328P.
- **Firmware & Low-Level Systems:** Embedded C, C++17/20, FreeRTOS, TinyML, Python, Assembly (ARM/x86), Linux Device Drivers.
- **Hardware Protocols & Buses:** SPI, I2C, UART/USART, CAN bus, PWM, ADC/DAC, DMA-driven waveform synthesis, GPIO edge interrupts.
- **RF, Wireless & IoT:** LoRa (SX1278 / 433–915MHz), Bluetooth Low Energy (BLE 5.0), Wi-Fi (ESP-NOW, HTTP, WebSockets), MQTT.
- **Analog Conditioning & Sensors:** OPA1612 Butterworth active filters, ACS712 Hall-effect current sensors, ZMPT101B voltage transducers, MPU6050 6-DoF IMUs, ultrasonic/piezo transducers, capacitive discharge spot welders.
- **DSP, Signal Processing & Edge AI:** Fast Fourier Transform (FFT) spectral shift analysis, Kalman filtering, Isolation Forest anomaly isolation, TFLite Micro.
- **Prototyping & Instrumentation:** KiCad PCB layout & schematic design, Digital Storage Oscilloscopes (DSOs), Logic Analyzers, JTAG/SWD.
- **Accessibility & Human Languages:** Certified in Indian Sign Language (ISL) fundamentals for non-verbal human-assistive interaction; English, Telugu, Hindi.

LEADERSHIP, COMPETITIONS & HONORS

- **IIT Madras Smart India Challenge – 2nd Runner Up & Youngest Panelist** *IIT Madras*
 - Awarded **2nd Runner Up** at IIT Madras Smart India Challenge for developing high-efficiency cyber-physical and systems architectures.
 - Honored as the **Youngest Panelist** at the summit, presenting on hardware miniaturization, sovereign edge diagnostics, and extreme economic engineering.
 - Showcased **Dust-Vision** (computer vision particulate tracking on GitHub) and **Dust-Sim** (real-time particle kinetics simulation live on the web), modeling particulate trajectories under dynamic thermal and wind vectors.
- **High School Autonomous Robotics AI Hackathon – 1st Place Winner** *11th Standard*
 - Won **1st Place** in an internal school AI robotics hackathon by engineering an autonomous self-parking robot from scratch.
 - Engineered a specialized optical line-following sensor array paired with a custom damping algorithm that completely eliminated the classic “snaking” oscillation issue.
 - Integrated global floor plan coordinate navigation utilizing the **A* search algorithm** for deterministic, collision-free slot parking.
- **Global Automotive Research Centre (GARC) Technical Delegation** *Chennai, India*
 - Selected technical delegate at the **GARC Chennai Meet**, engaging with automotive engineers on homologation standards and vehicle test beds.
 - Participated in high-level sessions on non-invasive vehicle telemetry, EV battery safety envelopes, and real-time sensor failure preemption.
- **National Cadet Corps (NCC) & Services Selection Board (SSB)** *Armed Forces Selection*
 - Trained cadet in the **National Cadet Corps (NCC)**, grounded in military field discipline, tactical leadership, and crisis management.
 - Recommended for and attended the **Services Selection Board (SSB) 3 times**, demonstrating high situational resilience and spatial awareness.

- **MotorSafe – Tri-Modal Edge-AI Predictive Maintenance Ecosystem | Public** *Link : Release v0.01*
 - Architected an edge-native predictive maintenance ecosystem monitoring industrial motors and preventing catastrophic bearing breakdowns.
 - Synchronized three high-frequency telemetry transducers: **ACS712** Hall-effect current (torque load), **ZMPT101B** (power quality), and **MPU6050** (mechanical vibrations).
 - Engineered on-chip Fast Fourier Transform (FFT) to extract spectral sidebands and deployed a localized unsupervised **Isolation Forest** on ESP32 to eliminate cloud latency and false alarms.
 - Streamed telemetry over MQTT into an interactive Three.js 3D digital twin dashboard reflecting motor health in real time.
- **AQUAPULSE – Cognitive Software-Defined Sonar Payload for AUVs | Public** *Link : Repository*
 - Engineered a cognitive, software-defined sonar payload for Autonomous Underwater Vehicles (AUVs) meeting the SIH26058 MoES/NIOT mandate.
 - Implemented zero-CPU DMA-to-DAC acoustic waveform synthesis on **STM32 / ESP32-S3** paired with active **OPA1612** Butterworth analog filtering.
 - Deployed an INT8 quantized **TinyML** edge policy engine to dynamically adapt ping frequency (100–480 kHz) across shifting thermoclines, cutting acoustic power consumption by **38%**.
- **LoRa-Connect – Offline Point-to-Point Vehicle Telemetry Link | Private** *Hardware Prototype | Private*
 - Developed a subscription-free, offline point-to-point LoRa telemetry system bridging smartphones directly to vehicles without cellular dependence.
 - Designed a custom USB-C RP2040 dongle establishing an encrypted 915MHz LoRa link to an onboard vehicle ESP32 node integrated into the CAN bus.
 - Achieved sub-mA standby current and bypassed mobile operating system hardware restrictions via USB CDC-NCM protocol encapsulation.
- **Smart Home Assistant Framework (NEXUS) – Local-First IoT Infrastructure | Private** *IoT Infrastructure | Private*
 - Architected a production-ready, containerized, local-first smart home ecosystem designed for absolute privacy and zero cloud dependence.
 - Integrated Home Assistant Core, ESPHome, Node-RED, and n8n orchestration inside a unified Docker Compose architecture.
 - Implemented native YAML sensor definitions for custom-built ESP32/ESP8266 ambient sensory nodes with local voice assistance via Wyoming.
- **SpecRAG – Datasheet-to-Firmware Validation Gateway | Public** *Link : Release v0.1.0*
 - Engineered an enterprise developer tool bridging 1,000-page silicon hardware datasheets with embedded C/C++ firmware.
 - Utilized the **Model Context Protocol (MCP)** and localized vector search to verify register bitmasks, clock-tree prescalers, and alternate-function pin muxing directly in the IDE prior to board fabrication.
- **EVolvAI – Physics-Informed EV Charging Infrastructure Engine | Public** *Link : Website | Release v1.0.0*
 - Formulated a research-grade cyber-physical framework coupling a **Physics-Constrained Generative Counterfactual VAE (GCD-VAE)** with a Multi-Objective Genetic Algorithm (GA).
 - Simulated distribution grid bus voltage drops, transformer thermal constraints, and climate stress anomalies to optimize city-scale charging node placement, achieving $R^2 \approx 0.5$ fidelity on sparse real-world data.

INTERDISCIPLINARY PLATFORMS & EDGE ENGINEERING PROJECTS

- **Microintegrated Flet Console – Embedded Low-Overhead GUI | Interdisciplinary** *Link : Website | Repository*
 - Designed an ultra-lightweight, cross-platform cyber-physical user interface combining microintegrated **Flet (Python + Flutter)** with embedded microcontrollers.
 - Engineered asynchronous event loops connecting serial UART/USB streams from embedded devices directly into real-time reactive widgets.
 - Delivered responsive 60 FPS hardware telemetry plotting and parameter calibration dashboards on low-spec host systems with zero web browser overhead.
- **errands-op (op-list) – Algorithmic Errand Routing Engine & VRPTW Solver | Public** *Link : Repository*
 - Built an offline-first algorithmic routing engine in **Flutter/Dart** solving multi-stop Vehicle Routing Problems with Time Windows (VRPTW) and bi-directional Home Assistant WebSocket synchronization.
 - Engineered 2D Kalman GPS filtering with dual-ring hysteresis geofencing (40m/80m) and learned in-store layout walk orders using Feedback Arc Set (FAS) cycle elimination and Kahn’s topological sort.
- **TurnBack (Activity-Mapper) – Local-First GPS Endurance Tracker | Public** *Link : Release v1.3.0*
 - Engineered an ultra-lightweight, battery-optimized GPS navigation app with a proprietary 54% fatigue-asymmetric turn-back engine.
 - Implemented stationary stop detection and Ramer-Douglas-Peucker (RDP) trajectory decimation, delivering 60 FPS rendering under < 5MB RAM.
- **SAFE-Inventory – Low-Power BLE Warehouse Asset Tracking Suite | Public** *Link : Website | Release v1.4.0*
 - Engineered a resilient warehouse asset management system featuring single-product batch scanning and set-and-forget Bluetooth LE hardware pairing.
 - Optimized transceiver duty-cycling on mobile and embedded scanners to provide multi-shift operational battery endurance.
- **EMT-Sims – Interactive 3D WebGL Electromagnetic Theory Suite | Public** *Link : Repository*
 - Engineered an interactive 3D WebGL physics simulation suite modeling Maxwell’s equations, Poynting vectors, and plane wave propagation through lossy dielectrics.
 - Implemented real-time boundary condition solvers computing Fresnel reflection coefficients and polarization transformations in the browser.

CERTIFICATIONS & TECHNICAL WORKSHOPS

- **Indian Sign Language (ISL) Certification** *Certified*
 - Earned formal certification in Indian Sign Language (ISL) fundamentals, enabling inclusive, non-verbal communication and human-assistive interface design.
- **Hardware Prototyping Workshop & Battery Spot Welder Fabrication** *Practical Engineering*
 - Designed and constructed a high-current capacitive discharge battery spot welder for precise nickel strip bonding and custom lithium-ion battery pack assembly.